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Measurely

Because baking belongs to everyone.

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Our Team

Lily

Product design
Fabrication
CAD

Sam

UX Design
Programming
Connectivity

Sophie

Electronics
Engineering
Mechanics


Vita

Statement of Intent

Often with age, people lose independence and the ability to care for themselves; baking goods is a task that becomes difficult. The heavy, repetitive, and tedious aspects of baking can be barriers, leading to frustration and a sense of loss when activities that feel central to one's identity are no longer feasible. We've seen this first hand with our own family members who have faced mobility and coordination issues with age and/or health problems.

Thus, **our project addresses the difficulty of measuring ingredients accurately and the disjointed nature of baking – where recipes, measuring cups, and scales are all separate, making the process harder to follow.**

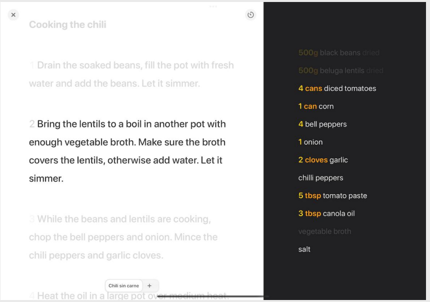
Elevator Pitch

Our project is an assistive kitchen device that makes baking easier for those with limited mobility. It's a countertop module that helps with liquid and dry measurements and leads users through various recipes with step-by-step instructions.

Precedents

Thermomix

Brava Oven



Mela App

Competitive Analysis

	Intuitive Interface	Accessible Interface	Mobile App/Website	Designed for Limited Mobility	Primary Function
Mela Recipe App	✅	✅	✅	❌	Guided recipes
Thermomix	✅	❌	✅	❌	Everything
Brava Oven	✅	❌	✅	✅	Oven

Audience Interviews

Goal:

Have conversations with people of different abilities and with different kitchen experiences/skill levels

Key findings:

- Precise measurements is a common pain point with baking, dry ingredients especially
- Measuring ingredients can be particularly difficult for those with mobility issues
- This device could be beneficial to those with varying levels of kitchen experience, skill, or ability

User Journey

1. Turn on the device and choose a recipe.
2. Set up bowls and get out the ingredients.
3. Add in larger dry ingredients.
4. Use assisted spice dispenser.
5. Use the assisted liquid hopper.
6. Combine the bowls and mix together.
7. Follow on screen and audible instructions to finish up.

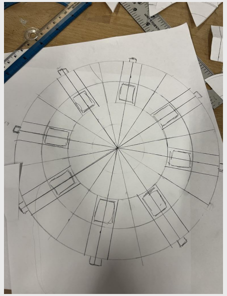
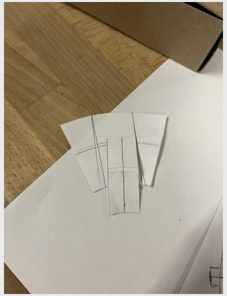
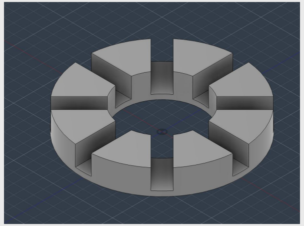
Feasibility test: spice dispensing mechanics

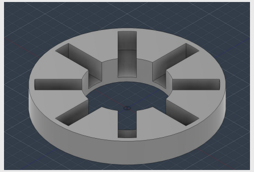
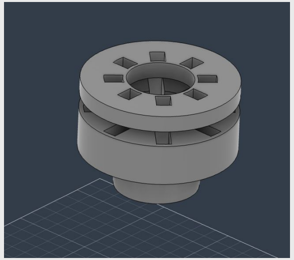
Goal:

Dispense dry ingredients via the smallest unit of measurement; inspired by a gumball machine

Process:

- Sketching
- First model
- Test Print
- Second model

Feasibility test: scale

Goal:

Accurately measure weight
(in grams) to prove scale feasibility

Process:

- 5kg load cell
- Printed spacers
- Laser cut wood → acrylic
- Calibration code

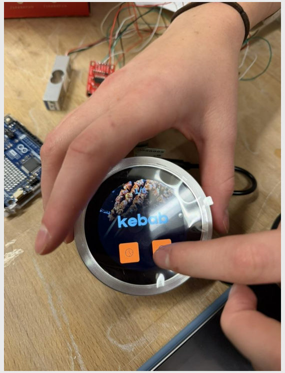
Feasibility test: interface

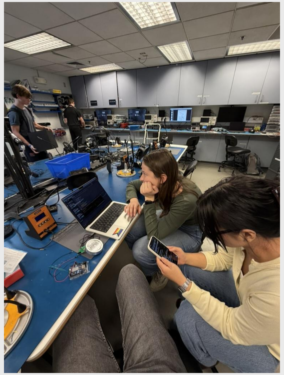
Goal:

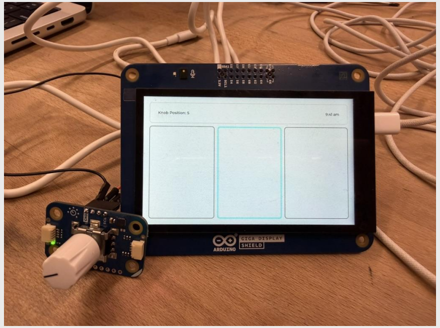
Create a basic, controllable interface with our desired components

Process:

- Experiment with many different components
- Decide on a pairing (knob and screen) that works for our needs
- Create test code to show feasibility







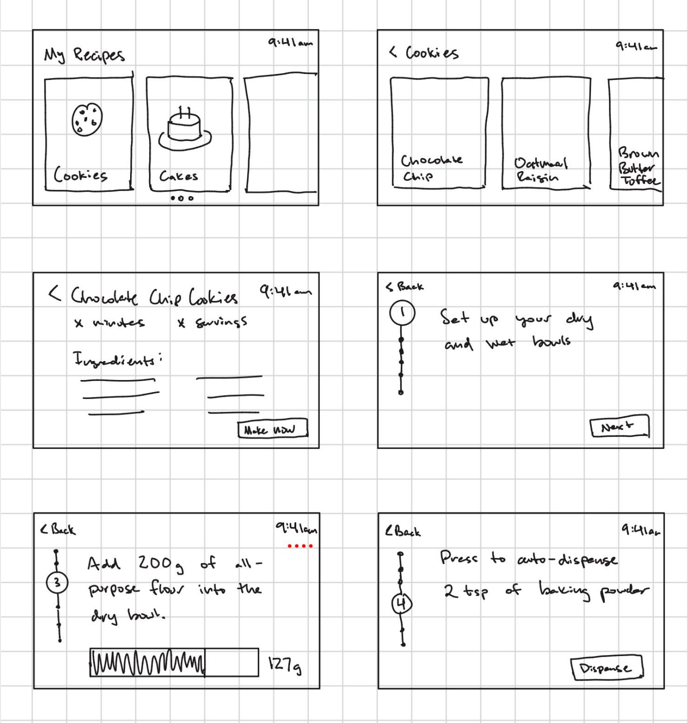
Feasibility test: design

Goal:

Establish a basic concept for home screen layout and recipe walkthrough

Process:

- Find inspiration from precedents and other technologies
- Determine necessary features
- Create rough, low-fidelity wireframes



Feasibility test: materials

Goal:

Determine which materials will be food safe, clean, and lightweight

Process:

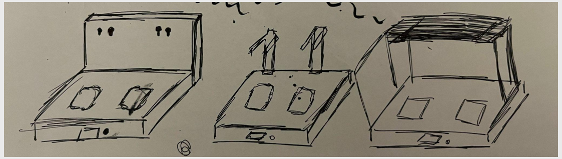
- Brainstorming
- Testing spills
- Cleaning
- Lifting

Dirty | Paper towel | Sponge

Project Renderings

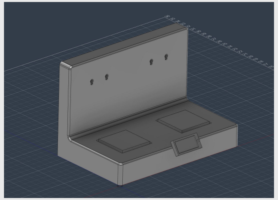
Goal:

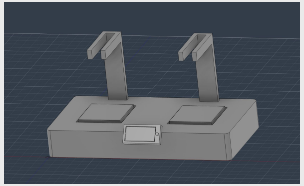
Determine design and layout

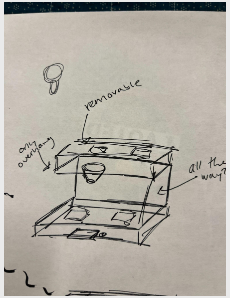


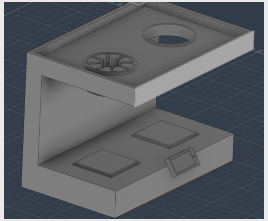
Process:

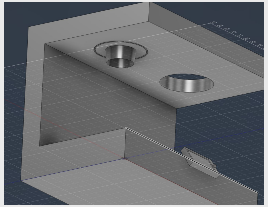
- Sketching
- First try
- Group feedback
- Adjustments

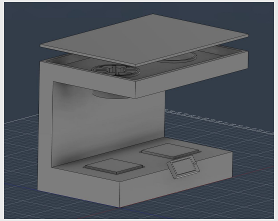




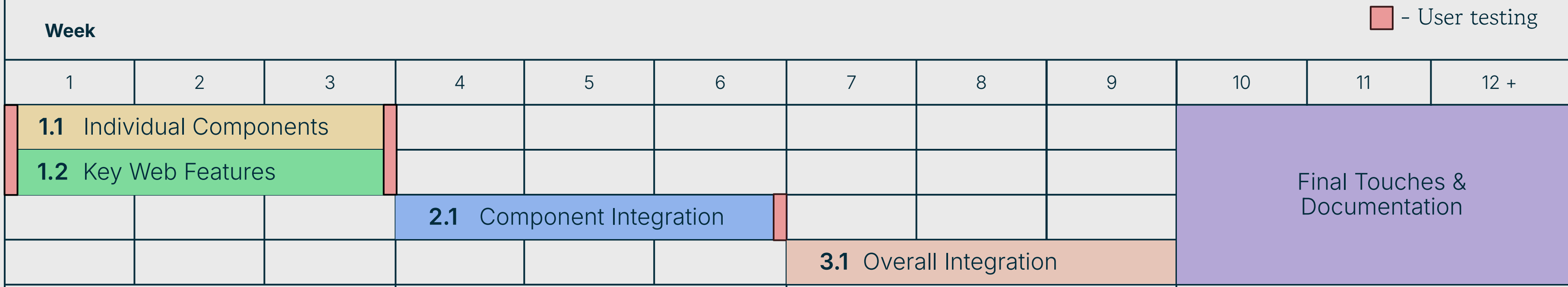








Implementation Plan & Timeline



Iteration 1

Scales
Spice Measurement
Liquid Measurement

Web app (for browsing and adding recipes)

Iteration 2

Integrate all physical components together in one designed enclosure

Connecting web app to physical device

Iteration 3

Full functionality (accurate measurements, recipe library with web app integration, complete step-by-step instructions)

Final project

Polished device

Product website and project documentation

Presentation @ EXPO

Thank you

We greatly appreciate your time and feedback!